Paper-based microfluidics (μPADs)

Group 3

Aaheli Das, Daniel Vega, Briseyda Duran Orozco, Christopher Gudiel-Gutierrez, Rana Hassan, Isabella Jimenez

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01

Significance

Significance of Paper-Based Microfluidics

Functionality

- Microfluidic paper-based analytical devices (μPADs)
- Low-cost, disposable platforms for diagnostics
- Depend on hydrophobic barriers (wax, marker, pencil) to control liquid flow

Research Focus

- H_0: Barrier type does not affect performance
- H_a:Barrier type does affect performance
- Performance criteria:
  - Wicking speed
  - Leakage area

Broader Significance

- Reproducibility → reliable results, prevents misdiagnosis
- Low-cost fabrication → accessible diagnostics, healthcare equity
- Illustrates role of statistical analysis in biomedical research

02

Proposed Methods

Objective

- Fabricate 40 μPAD strips (60 × 20 mm)
- Test three barrier types: Wax, Marker, Pencil
- Include positive (with barriers) and negative (no barriers) controls
- n = 9 replicates per condition (randomized order)
- Record temperature (22–24 °C) and humidity (40–60%)
- Measure Wicking Speed and Leakage Area

Experimental Design Controls

- Cut all coupons from the same chromatography sheet when possible
- Use smartphone imaging with fixed exposure
- Focus analysis on channel region only
- Measure:
  - Wicking Speed
  - Leakage Area

Materials

The materials needed for this experiment include;

- Whatman chromatography paper
- Wax crayon
- Hot Plate (150 °C)
- Sharpie (fine-tip, water-resistant)
- HB or 2B pencil
- Food dye (for visualization)
- Smartphone (for imaging)
- ImageJ software + laptop

Barrier Fabrication

- Wax: Draw → heat at 150 °C for 2–4 min (full penetration)
- Marker: Draw → air dry for 5–30 min (solvent evaporation)
- Pencil: 5–10 strokes (optional light burnish for density)
- Combinations: Wax+Marker, Wax+Pencil, Marker+Pencil, All three
- 5 replicates per type

Testing Procedure

Two responses will be measured to evaluate barrier performance:

- **Wicking Speed:**
  - Add 10 μL dyed water to inlet
  - Record 60 s video
  - Analyze distance vs. time in ImageJ → compute speed
- **Leakage Area (%):**
  - Capture top-view photo
  - Quantify dye spread beyond 3 mm channel in ImageJ

03

DoE Table

DOE Overview

- DoE Type: Comparative design using independent μPAD coupons
- Primary factor: Barrier type (Wax, Marker, Pencil)
- Response variables: Wicking speed and leakage area
- Experimental unit: One coupon per barrier condition
- Controls: One positive control (wax double barrier) and one negative control (no barrier) per batch

Not screening:
- Only one primary categorical factor (barrier type)
- Screening would require testing multiple additional variables (paper type, channel width)

Not a response surface model:
- No continuous independent variables (e.g., reagent concentrations)
- Focused on comparing discrete barrier materials

Not a regression study:
- Core analysis based on group comparison (ANOVA), not correlation or slope fitting

DOE Table

ChipID	BarrierType	SampleCondition	Replicate	RandomOrder	Avg. Wicking Speed	Avg. Leakage Area %	Temp. (C)	Humid (%RH)	Date	Time	Notes
C001	Wax	Flow (Dye)	1	17							
C002	Marker	Flow (Dye)	1	42							
C003	Pencil	Flow (Dye)	1	5							
C004	Wax	Flow (Dye)	2	33							
C005	Marker	Flow (Dye)	2	2							
C006	Pencil	Flow (Dye)	2	55							
C007	Wax	Flow (Dye)	3	51							
C008	Marker	Flow (Dye)	3	31							
C009	Pencil	Flow (Dye)	3	47							
PC1	Positive Control	Wax double	1	11							
NC1	Negative Control	No Barrier	1	9							
C010	Wax	Flow (Dye)	4	62							
C011	Marker	Flow (Dye)	4	6							
C012	Pencil	Flow (Dye)	4	50							
C013	Wax	Flow (Dye)	5	29							
C014	Marker	Flow (Dye)	5	39							
C018	Pencil	Flow (Dye)	5	3							
C019	Wax	Flow (Dye)	6	44							
C020	Marker	Flow (Dye)	6	19							
C021	Pencil	Flow (Dye)	6	26							
PC2	Positive Control	Wax double	2	20							
NC2	Negative Control	No Barrier	2	14							
C022	Wax	Flow (Dye)	7	24							
C023	Marker	Flow (Dye)	7	31							
C024	Pencil	Flow (Dye)	7	87							
C025	Wax	Flow (Dye)	8	1							
C026	Marker	Flow (Dye)	8	10							
C027	Pencil	Flow (Dye)	8	38							
C028	Wax	Flow (Dye)	9	67							
C029	Marker	Flow (Dye)	9	72							
C030	Pencil	Flow (Dye)	9	8							
PC3	Positive Control	Wax double	3	63							
NC3	Negative Control	No Barrier	3	92							

All chips will be run on same day, same time

⅓ DOE Table

3 replicates per batch, 9 total

Randomized runs, eliminate sequence bias

Temperature and Humidity- Documented but not used for Stats Analysis

Experimental Chips = 27 Total = 33

ChipID	BarrierType	SampleCondition	Replicate	RandomOrder	Avg. Wicking Speed	Avg. Leakage Area %	Temp. (C)	Humid (%RH)	Date	Time	Notes
C001	Wax	Flow (Dye)	1	17							
C002	Marker	Flow (Dye)	1	42							
C003	Pencil	Flow (Dye)	1	5							
C004	Wax	Flow (Dye)	2	33							
C005	Marker	Flow (Dye)	2	2							
C006	Pencil	Flow (Dye)	2	55							
C007	Wax	Flow (Dye)	3	51							
C008	Marker	Flow (Dye)	3	31							
C009	Pencil	Flow (Dye)	3	47							
PC1	Positive Control	Wax double	1	11							
NC1	Negative Control	No Barrier	1	9							

One (+), one (-) per batch 6 controls

Dependent variables Avg Wick Speed + Leakage Area

[4,5,6]

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Expected Results

Statistical Analysis

- Data Testing:
  - Shapiro-Wilk Test for normality
    - P-value > 0.05 → normally distributed
  - If data is normal: Perform one-way ANOVA (Wicking Speed and Leakage %)
    - Statistical Significance
      - P value<0.05: Reject null hypothesis
        - H_0: $\mu_W = \mu_M = \mu_P$
        - H_a: $\mu_W \neq \mu_M \neq \mu_P$

$$W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Squares (MS)	F
Within	$SSW = \sum_{j=1}^{k}\sum_{j=1}^{l}(x-\bar{x}_j)^2$	$df_w = k-1$	$MSW = \frac{SSW}{df_w}$	$F = \frac{MSB}{MSW}$
Between	$SSB = \sum_{j=1}^{k}(\bar{x}_j - \bar{x})^2$	$df_b = n-k$	$MSB = \frac{SSB}{df_b}$	
Total	$SST = \sum_{j=1}^{n}(\bar{x}_j - \bar{x})^2$	$df_t = n-1$		

    - Significant difference: Tukey's HSD (Honestly Significant Difference)

$$T = q_{\alpha(c,n-c)}\sqrt{\frac{MSE}{n_i}}$$

Statistical Analysis

- If not normal: Use non-parametric tests
  - Kruskal-Wallis test
    - P < 0.05 → statistically significant
    - Significant difference between the groups
      - Dunn's test - determine which specific pairs of groups are significantly different from each other.

$$H = \frac{12}{n(n+1)} \sum \frac{R_i^2}{n_i} - 3(n+1)$$

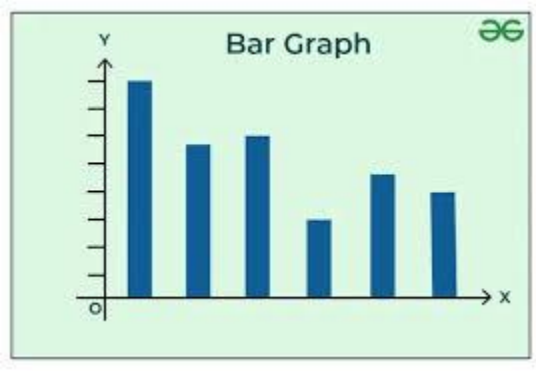
$$z_{A,B} = \frac{R_A - R_B}{\sqrt{\frac{N(N+1)}{12}\left(\frac{1}{n_A} + \frac{1}{n_B}\right)}}$$

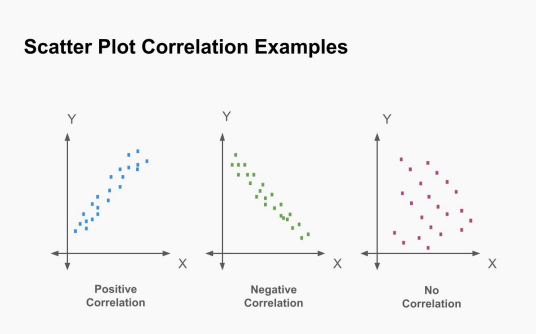
Graphing Results

Wicking Speed vs. Barrier Type: Bar graph
- Comparing: Wax, Marker, and Pencil barrier
- Average wicking speed (e.g., distance traveled over time)

Leakage Area vs. Barrier Type: Scatter plot
- Fluid leakage area for each barrier type
- Show spread and outliers

Leakage Area vs. Wicking Speed: Scatter plot + linear regression line
- X-axis (independent variable): wicking speed
- Y-axis (dependent variable): leakage area





05

Time Budget

Weeks	Dates	Topics	Time Estimated
1	10/20-10/24	DOE Project Proposal Final Submission	~2-3 hours
2	10/27-10/31	Receive Material/ Barrier Fabrication	~1-2 hours per day, around 4 days at most, n=33, (Wax, Marker, Pencil)
3	11/3-11/7	Experimental Runs	Complete in one day, total 33 runs, Split workload among group (groups of 2, each run 11 runs in one day)
4-5	11/10-11/21	Image Analysis/Buffer Time	1 week to analyze all runs, another week for buffer time if needed
5	11/17-11/21	Data Organization and Statistics	3-5 days, data cleaning and plotting, Use statistics to draw conclusions
6	11/24-11/28	Start Final DoE Slides and Project Paper/ Buffer Time	3-5 days, 2~3 hours, Focus on discussion portion and drawing conclusions
7	12/1-12/5	Final DoE Slides and Project Video	~2-3 hours for the slides and the video, can be split into three days if needed
8	12/8	Final DoE Project Report Due	~3-5 days (Have been working on Report so this is the final touch up)

Objective & Significance

Objective:

We compare wax, marker, and pencil hydrophobic barriers. Barrier performance defined by wicking speed and leakage area

Significance:

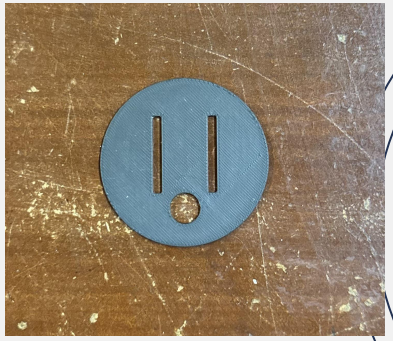
Barrier quality affects reproducibility, accuracy, and risk of misdiagnosis. Understanding fabrication methods improves device reliability. µPADs enable low-cost, point-of-care diagnostic

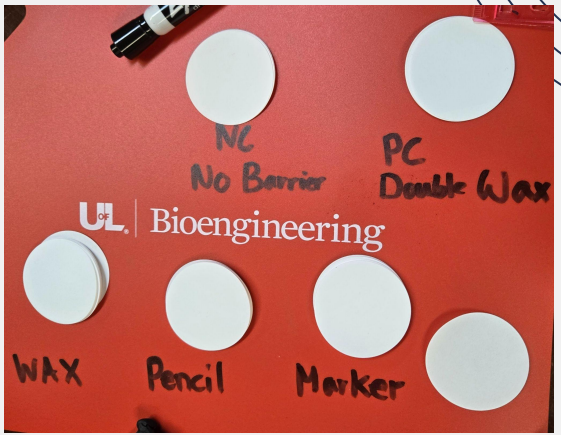
Methods: Barrier Fabrication

- Barriers drawn with wax / marker / pencil using 3D-printed stencil template
- Circular paper device (55 mm) with
- stencil-drawn barriers
- 10 µL dye pipetted into center

Variables measured:

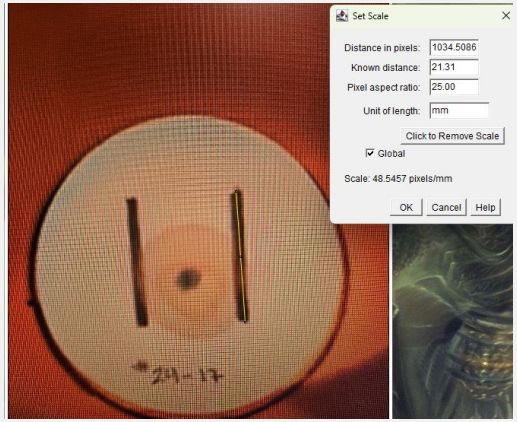
- Wicking speed (time → barrier contact)
- Leakage area (ImageJ)

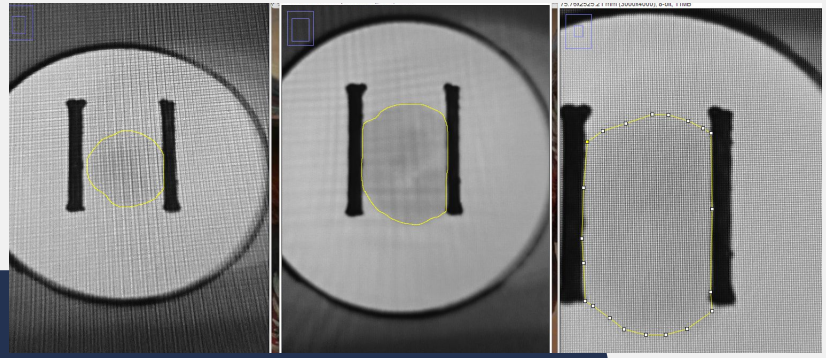




Experimental Setup

- Randomized run order (DoE table)
- Controlled environment logging
- Video-based measurement
- ImageJ quantification





Observations:

Leakage Area:

- Wax = lowest leakage
- Pencil = highest leakage

Tukey HSD			
Group	Mean	Variance	n
Wax	50.77	248.67	9
Marker	64.47	212.32	9
Pencil	27.26	123.55	9

MSE	194.85
qcrit	3.51
SE	4.65
HSD	16.31

Anova: Single Factor

You reject the null hypothesis:

SUMMARY				
Groups	*Count*	*Sum*	*Average*	*Variance*
Wax	9	456.93	50.77	248.67
Marker	9	580.26	64.47	212.32
Pencil	9	245.35	27.26	123.55

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	6375.84	2	3187.92	16.36	3.29E-05	3.40
Within Groups	4676.35	24	194.85			
Total	11052.19	26				

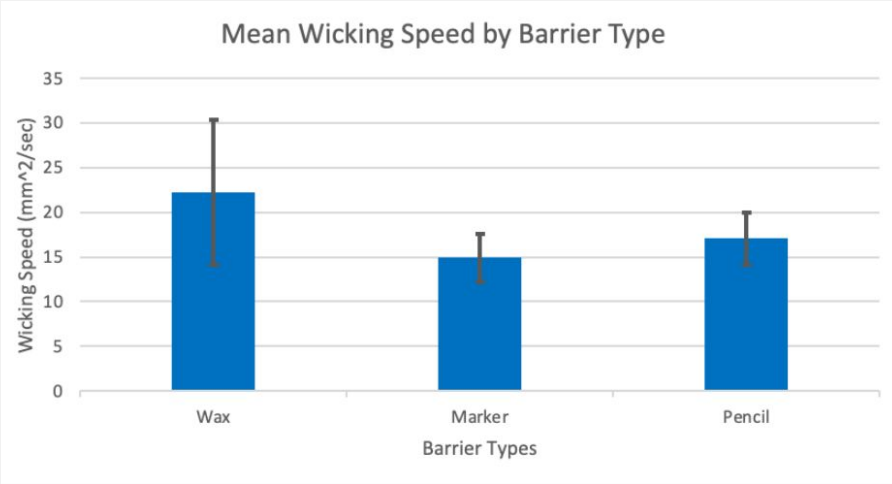
16.361 > 3.4028

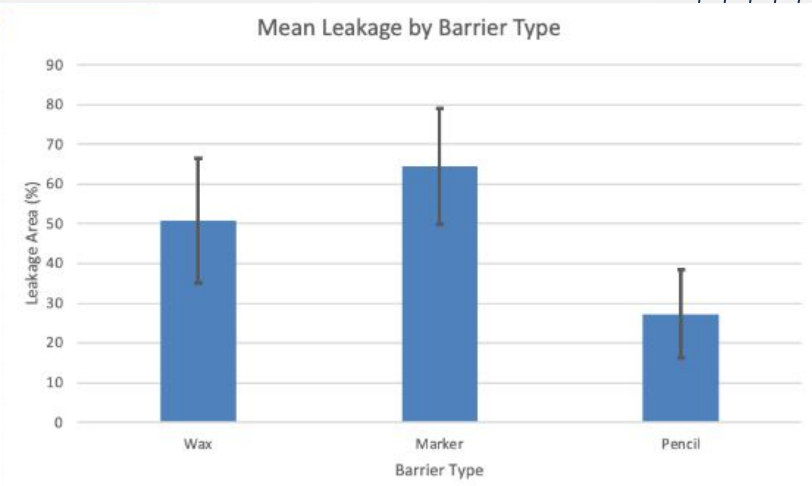
This means the variation between barrier types is far greater than the variation within each barrier type

P-value You reject the null hypothesis.
Barrier type significantly affects leakage area $p < 0.05$

Comparison	Difference	Significant?
Wax vs Marker	13.70	Not significant
Wax vs Pencil	23.51	Significant
Marker vs Pencil	37.21	Significant

Graphs


Mean Wicking Speed by Barrier Type
Wicking Speed (mm^2/sec)
35
30
25
20
15
10
5
0
Wax
Marker
Pencil
Barrier Types


Mean Leakage by Barrier Type
Leakage Area (%)
90
80
70
60
50
40
30
20
10
0
Wax
Marker
Pencil
Barrier Type

Observations

Wicking Speed:

- Wax = slowest flow

Tukey HSD			
Group	Mean	Variance	n
Wax	22.22	65.93	9
Marker	14.90	7.05	9
Pencil	17.06	8.84	9
MSE	27.27		
qcrit	3.51		
SE	1.74		
HSD	6.10		

Anova: Single Factor

You reject the null hypothesis:

SUMMARY				
Groups	*Count*	*Sum*	*Average*	*Variance*
Wax	9	199.98	22.22	65.93
Marker	9	134.10	14.90	7.05
Pencil	9	153.55	17.06	8.84

ANOVA						
Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	254.60	2	127.30	4.67	1.94E-02	3.40
Within Groups	654.56	24	27.27			
Total	909.16	26				

4.60 >3.40

There is a statistcally significant difference in wicking speed between at least two of the the barrier types

p=0.0194 < 0.05, reject the null

Comparison	Difference	Significant?
Wax vs Marker	7.32	Significant
Wax vs Pencil	5.16	Not Significant
Marker vs Penci	2.16	Not Significant

Resources

1. Dungchai, W., Chailapakul, O., & Henry, C. S. (2011). A low-cost, simple, and rapid fabrication method for paper-based microfluidics using wax screen-printing. Analyst, 136(1), 77-82. https://doi.org/10.1039/C0AN00406E
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6. Lee, M., et al. (2018). Fabrication of miniaturized paper-based microfluidic devices (µPADs). *Cellulose*, *25*(…), 387–395.

Acknowledgements

Tasks	**Name:**
Proposed Methods, DoE Table	Aaheli Das
Expected Results	Briseyda Duran Orozco
Proposed Methods	Rana Hassan
Time Budget	Christopher Gudiel-Gutierrez
Proposed Methods, DoE Table	Isabella Jimenez
Significance	Daniel Vega

Thanks!

Do you have any questions?